

Network Security 1 Project Rubric 7

Monitor and Respond to Network Security Events

Submit a report detailing the monitoring of network security events, including at least 1 security incident identified and the steps taken for incident response, supported by logs and screenshots.

Monitor and Respond to Network Security Events

1. Monitoring Network Security Events

Network monitoring involves continuously observing network activity to identify suspicious or unauthorized behavior. In a LAN, administrators can use several tools and techniques:

- **Firewall logs** – Record blocked and permitted network connections.
- **Intrusion Detection Systems (IDS)** – Detect suspicious traffic and potential attacks.
- **Antivirus/Endpoint Protection** – Identifies malware and other threats on computers.
- **Authentication logs** – Record successful and failed login attempts.
- **Network monitoring software** – Tracks unusual traffic, device failures, and performance problems.
- **Security Information and Event Management (SIEM)** – Collects and analyzes security logs from multiple devices in one location.

2. Examples of Security Events

Security Event	Possible Indication	Recommended Response
Multiple failed logins	Password attack	Lock the account and investigate
Unknown device connects	Unauthorized access	Block the device and identify it
Unusual data transfer	Possible data theft	Isolate the affected device
Malware detected	Infected computer	Disconnect and scan the device
Port scanning	Reconnaissance/attack attempt	Block the source and investigate

Firewall repeatedly blocks
traffic

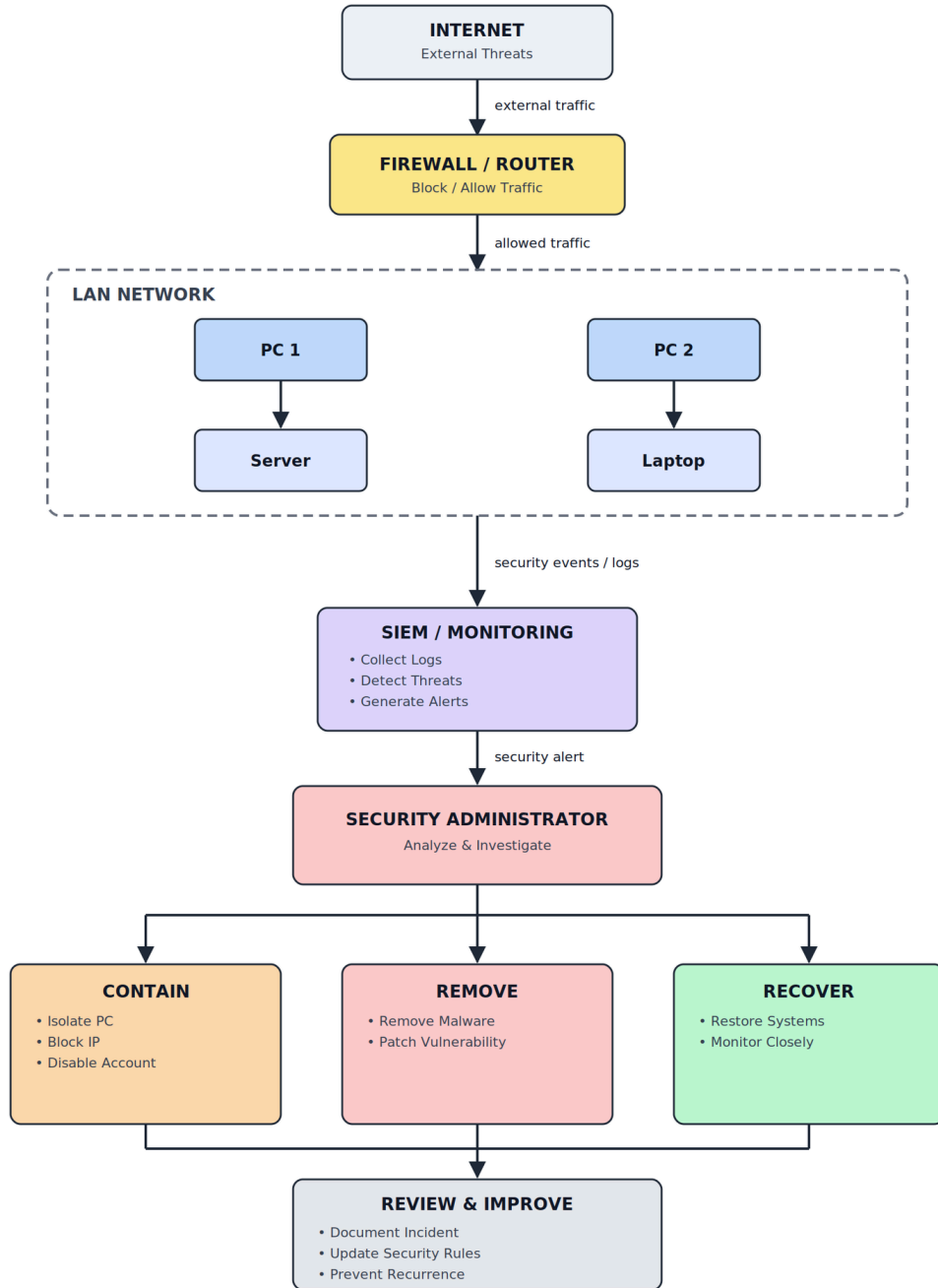
Possible attack

Review logs and source
addresses

3. Responding to a Security Event

A simple incident-response process can be followed:

Responding to a Security Event — Incident Response Workflow



Arrows follow the sequence of the incident-response workflow, from detection through recovery.

4. Example

Suppose an administrator notices **hundreds of failed login attempts** against a server.

1. The monitoring system generates an alert.
2. The administrator checks the logs to determine the source and timing.
3. The suspicious source is blocked using the firewall.
4. The affected account can be temporarily disabled or have its password reset.
5. The server is checked for signs of unauthorized access.
6. The incident is documented and security rules are updated if necessary.

5. Importance

Monitoring and responding to network security events helps an organization **detect attacks early, limit damage, protect sensitive information, and maintain network availability**.

Regularly reviewing security logs and having a clear incident-response procedure allows administrators to respond quickly rather than waiting until a security breach becomes serious.